

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 08



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic

Mixtures Alligations



Topics to be Covered



Topic-1

Number System



[MCQ]



#Q. How many litres of water must be added to 16 litres milk containing 10% water to dilute it with 20% water?

Assignment

- A 3 litres
- B 2 litres**
- C 1 litre
- D 4 litres

Mix Water

10% 100%

Mix : water = 80 : 10

= 8 : 1

= 16 : 2

80 10

~~Mix > 18/1~~

~~Water > 16 + 2 = 18~~

~~$\frac{20}{18} \times 100 = 20\%$~~

[MCQ]



#Q. A mixture contains milk and water in the ratio 5:1. On adding 5 liters of water, the ratio of milk to water becomes 5:2. the quantity of milk in the mixture earlier was:

Assignment

- A 16 litres
- B 25 litres**
- C 32.5 litres
- D 22.75 litres

Milk

$$\frac{7}{42} = \frac{1}{6}$$
$$\frac{5}{x} \times \frac{5}{30} = \frac{25}{lit}$$

Water

$$1 = \frac{42}{42}$$

Milk : water

$$= 30 : 5$$
$$6 : 1$$
$$= 30 : 5$$

$$\frac{30}{42}$$

$$\frac{5}{42}$$

$$\frac{2}{7} = \frac{12}{42}$$

[MCQ]



#Q. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

Assignment

A 3:8

B 5:3

C 7:3

D 4:5

$$\begin{array}{cc} A & B \\ 15 & 20 \\ & \swarrow \searrow \\ & 16.5 \\ & \swarrow \searrow \\ 3.5 & 1.5 \end{array}$$
$$A:B = 3.5 : 1.5$$
$$= 35 : 15$$
$$= 7 : 3$$

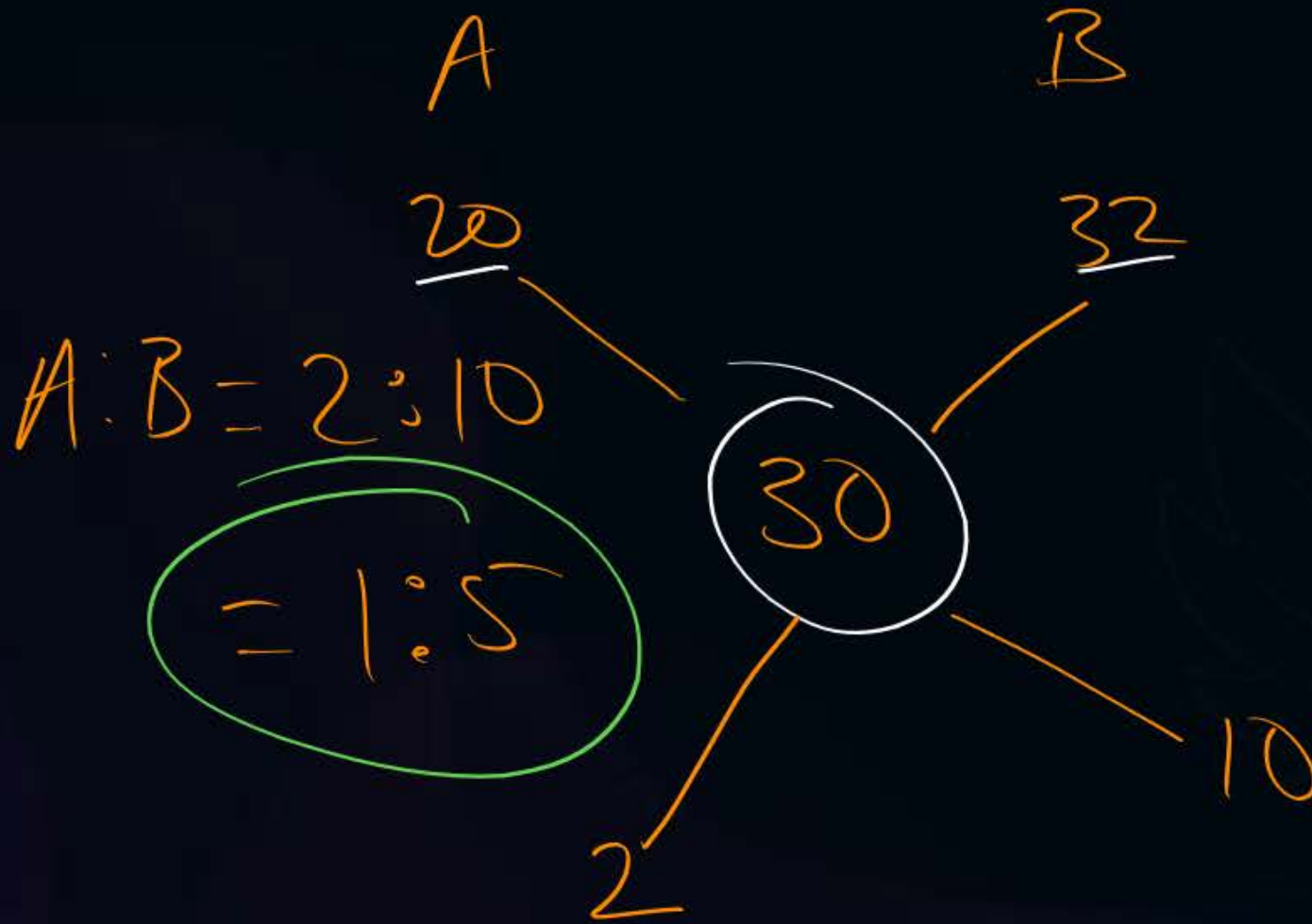
[MCQ]



#Q. In what ratio must Ankit mix two varieties of sugar worth ₹20 per kg and ₹32 per kg so that by selling the mixture at ₹36 per kg he may gain 20%?

Assignment

- ☒ A 1:5
- ☐ B 3:5
- ☐ C 2:3
- ☐ D 3:2



$$SP = 36$$

$$P\% = 20\%$$

$$\frac{36}{1.2} = 1.2$$

$$\frac{36}{1.2} = CP = 30$$

Basics of Number System

' ∞ '

Natural Number

$$5 \overline{) 15} (3$$

$$\frac{15}{5} = 3$$

$$\frac{16}{2} = 8$$

$$\frac{0}{9} = 0$$

$$2 \overline{) 16} (8$$

$$9 \overline{) 0} (0$$

$$0 \overline{) 5} ($$

$$\frac{5}{0} = \infty$$

1 ① Rational $\rightarrow \frac{p}{q}$
2 3 Prime No.
2 ONLY Two factors

Non-Prime
 OR
 Composite

1 $\rightarrow$

? factor

6 $\rightarrow$ 1, 2, 3, 6

? Multiples 6 $\rightarrow$ 6, 12, 18, 24, 30, 36, 42

?

?

48, 54, 60, 66, 72, 78, 84
 90, 96, 102, ... ∞

$$\underline{42.\overline{4}} = \frac{382}{9}$$

$$3 = \frac{3}{1}$$

$$-8 = \frac{-8}{1}$$

$$4.6 = \frac{46}{10}$$

$$0 = \frac{0}{1}$$

Irrational?

$\sqrt{2} = 1.41421356298734732186434971\dots$

Irrational

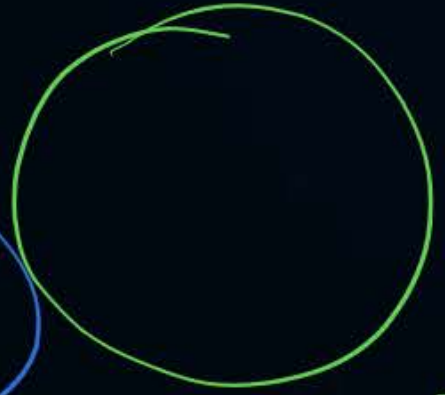
Decimals

Terminating

Non-Terminating

Recurring

Non-Recurring



$2\pi R$

Irrational
 π

$\frac{22}{7}$

3.14

Rational

FRIDAY
 1st Sep, 2000

S

1999

→ 1900 +

99 → 4

L: 4
 24
 75

48

75

6 + 5 = 11

J - 3
 F - 1
 M - 3
 A - 2
 M - 3
 J - 2

J - 3
 A - 3
 S - 1
 0

$$7 \overline{) 22} (3 \cdot 7 \overline{) 28571}$$

$$\begin{array}{r} 60 \\ 56 \\ \hline 40 \\ 35 \\ \hline 50 \\ 49 \\ \hline 10 \\ \hline \end{array}$$

$$\begin{array}{r} 10 \} \\ 7 \} \\ \hline 30 \\ 28 \\ \hline 20 \\ 14 \\ \hline 60 \end{array}$$

[MCQ]



#Q. Find the answer of given expression:

$$\underbrace{1-2}_{-1} + \underbrace{3-4}_{-1} + \underbrace{5-6}_{-1} \dots \dots \dots 2025.$$

2024

$$\begin{array}{r} + 2025 \\ - 1012 \\ \hline \end{array}$$

$$\underline{\underline{1013}}$$

$$\underline{\underline{1013}}$$



[MCQ]



#Q. What would the unit digit in the answer of given expression:

$$1! + 2! + 3! + \dots + 2025!$$

Assignment

[MCQ]



#Q. What would the remainder when the answer of given expression is divided by 3?

$$1! + 2! + 3! + \dots + 2025!$$

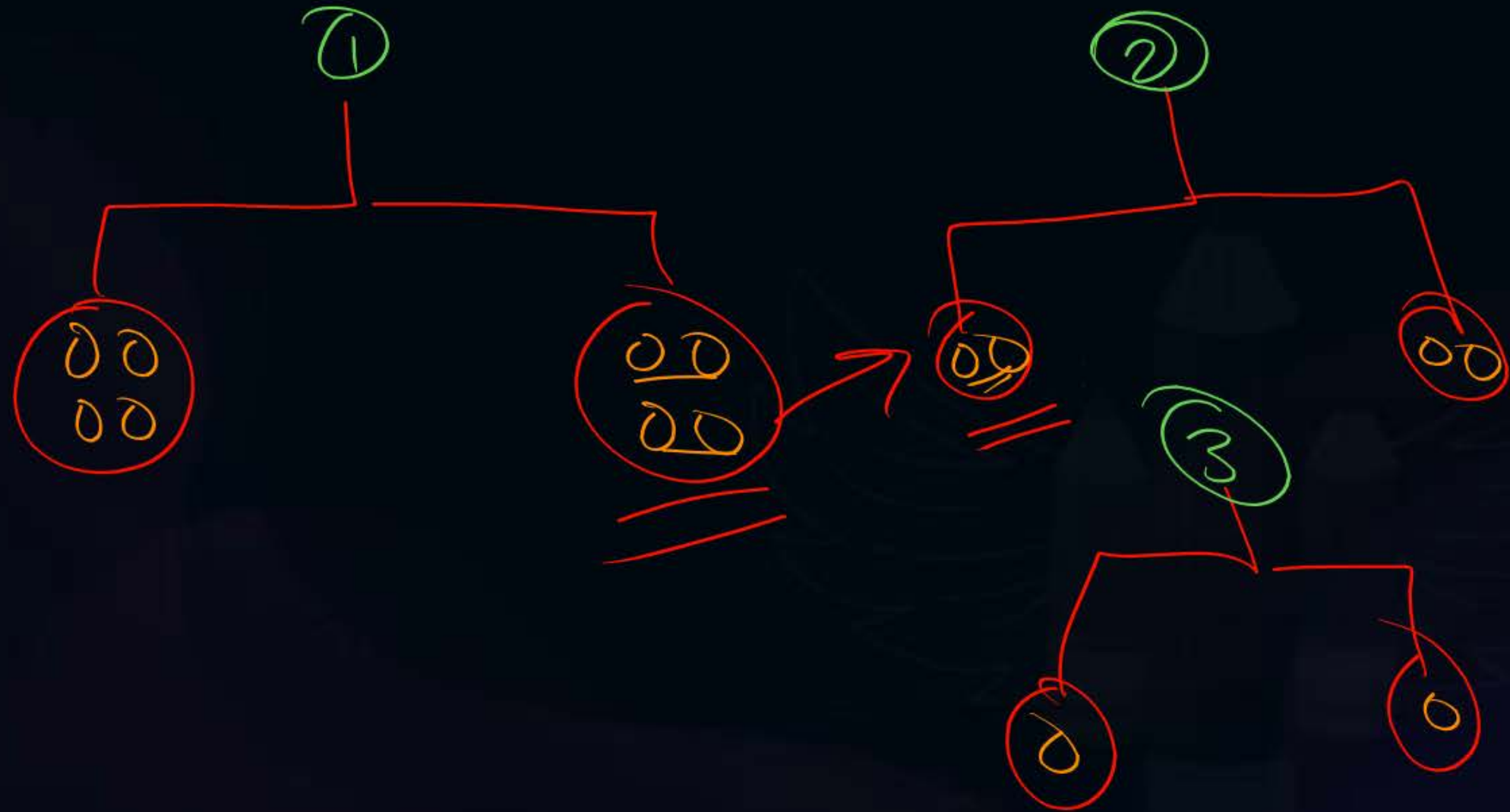
Assignment

[MCQ]

2 chance



#Q. There are 8 balls of which one is defective. Given that the defective ball is of less weight and remaining are of equal weights. What are the minimum number of chances a common balance is to be used to find the defective one?



[MCQ]



#Q. A bag consist of 48 red colour balls, 16 green colour balls, 12 yellow colour balls, 11 black colour balls and 11 white colour balls. How many minimum number of balls are to be taken out from the bag randomly so that we get at least two balls of same colour?

at least

(blind fold)

6 balls

[MCQ]



#Q. A shelf consist of 40 socks. 40% of these are black and remaining are white. How many minimum number of socks are to taken out from the shelf randomly (blindly), so that we get atleast two white socks?

18 $\Rightarrow$ 2 white

40
(B) Black white (W)
16 24

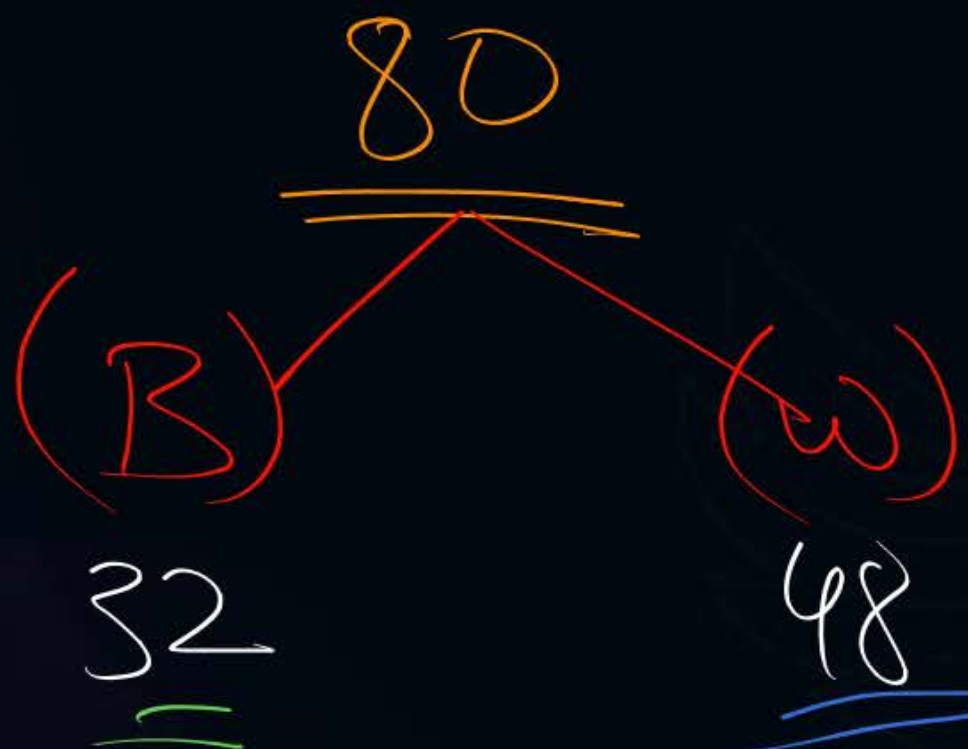
2 black ?
26

[MCQ]



#Q. A box consist of 40 pairs of shoes of equal size. 40% of these are black and remaining are white. How many minimum number of shoes are to taken out from the box randomly, so that we get at least a pair of black shoes?

$$\begin{array}{r} 48 \\ + 16 \\ + 1 \\ \hline 65 \end{array}$$



at least 2 black?

~~50~~

[MCQ]



#Q. How many zeroes would be at the end in the answer of $50!$?

Assignment

- A** 12
- B** 15
- C** 23
- D** 24



2 mins Summary



Topic

Number System - 01



THANK - YOU